ASSIGNMENT

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Branch: BE-CSE **Section/Group:** KRG-3B
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Subject Name: Full Stack-II **Subject Code:** 23CSH-309

1. Aim:

To design and implement a structured frontend module of the EcoTrack application using modern React practices, Vite tooling, React Router for navigation, Material UI for styling, and performance optimization techniques such as React.memo and useCallback.

2. Objective:-

- To understand the fundamentals of React functional components and Vite project setup.
- To implement client-side routing using React Router with protected routes.
- To manage state effectively using useState and useEffect.
- To implement localStorage for persistent data storage.
- To integrate external APIs using asynchronous fetch requests.
- To implement loading and error handling mechanisms during API calls.
- To optimize component rendering using React.memo and useCallback.
- To design a clean and professional dark-themed UI using Material UI.
- To maintain proper project structure using components and pages separation.

3. . Tools & Technologies Used

- React (Functional Components)
- Vite
- React Router DOM
- Material UI (MUI)
- JavaScript ES6+
- Browser LocalStorage
- REST API Integration

4. Implementation/Code:

COUNTER-DISPLAY:

```
import React from "react";
import { Typography } from "@mui/material";

const CounterDisplay = ({ count, goal }) => {
  console.log("CounterDisplay Rendered");

  return (
    <>
      <Typography variant="h5" gutterBottom>
        {count} / {goal} glasses completed
      </Typography>

      {count >= goal && (
        <Typography color="success.main" fontWeight="bold">
          Goal Reached 🎉
        </Typography>
      )}
    </>
  );
};

export default React.memo(CounterDisplay);
```

NAVBAR:

```jsx
import { AppBar, Toolbar, Button, Typography, Box } from "@mui/material";
import { useNavigate } from "react-router-dom";

const Navbar = () => {
  const navigate = useNavigate();
  const token = localStorage.getItem("token");

  const handleLogout = () => {
    localStorage.removeItem("token");
    navigate("/login");
  };

  if (!token) return null;

  return (
    <AppBar position="static" elevation={1}>
      <Toolbar sx={{ display: "flex", justifyContent: "space-between" }}>
        <Typography variant="h6">
          EcoTrack Dashboard
        </Typography>

        <Box>
          <Button color="inherit" onClick={() => navigate("/dashboard")}>
            Dashboard
          </Button>

          <Button color="inherit" onClick={() => navigate("/dashboard/water")}>
            Water Tracker
          </Button>

          <Button color="error" onClick={handleLogout}>
            Logout
          </Button>
        </Box>
      </Toolbar>
    </AppBar>
  );
};

export default Navbar;
```

PROTECTEDROUTE.jsx:

```jsx
import { Navigate } from "react-router-dom";

const ProtectedRoute = ({ children }) => {
  const token = localStorage.getItem("token");
  return token ? children : <Navigate to="/login" />;
};

export default ProtectedRoute;
```

DASHBOARD:

```jsx
import { Container, Typography, Card, CardContent } from "@mui/material";

const Dashboard = () => {
  return (
    <Container maxWidth="md" sx={{ mt: 6 }}>
      <Card elevation={3}>
        <CardContent>
          <Typography variant="h4" gutterBottom>
            Welcome to EcoTrack
          </Typography>

          <Typography variant="body1" color="text.secondary">
            This dashboard allows users to track their daily water intake,
            set hydration goals, and receive health tips.
          </Typography>
        </CardContent>
      </Card>
    </Container>
  );
};

export default Dashboard;
```

APP.jsx:

```jsx
import { Routes, Route, Navigate } from "react-router-dom";
import Login from "./pages/Login";
import Dashboard from "./pages/Dashboard";
import WaterTracker from "./pages/WaterTracker";
import ProtectedRoute from "./components/ProtectedRoute";
import Navbar from "./components/Navbar";

function App() {
  return (
    <>
      <Navbar />
      <Routes>
        <Route path="/login" element={<Login />} />

        <Route
          path="/dashboard"
          element={
            <ProtectedRoute>
              <Dashboard />
            </ProtectedRoute>
          }
        />

        <Route
          path="/dashboard/water"
          element={
            <ProtectedRoute>
              <WaterTracker />
            </ProtectedRoute>
          }
        />

        <Route path="*" element={<Navigate to="/login" />} />
      </Routes>
    </>
  );
}

export default App;
```

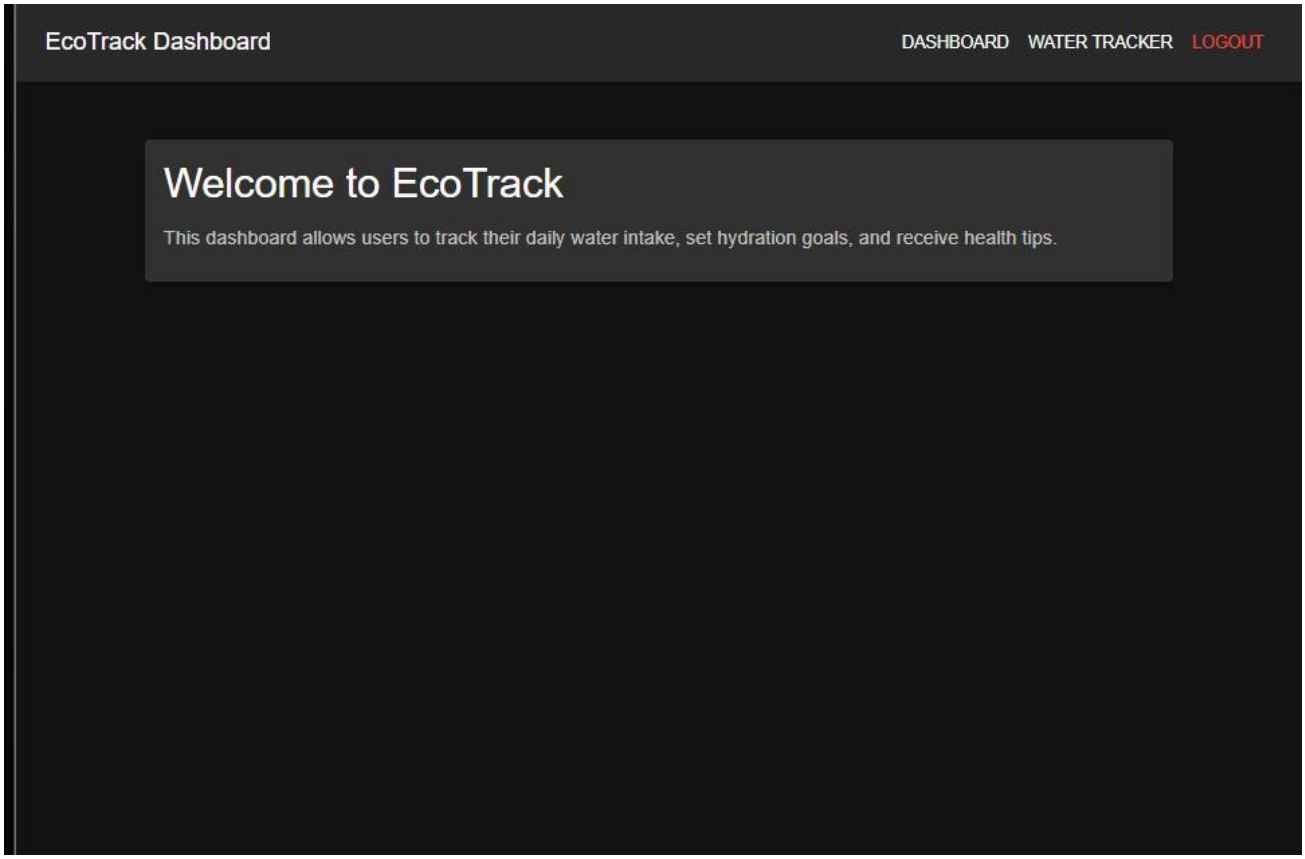
MAIN.jsx:

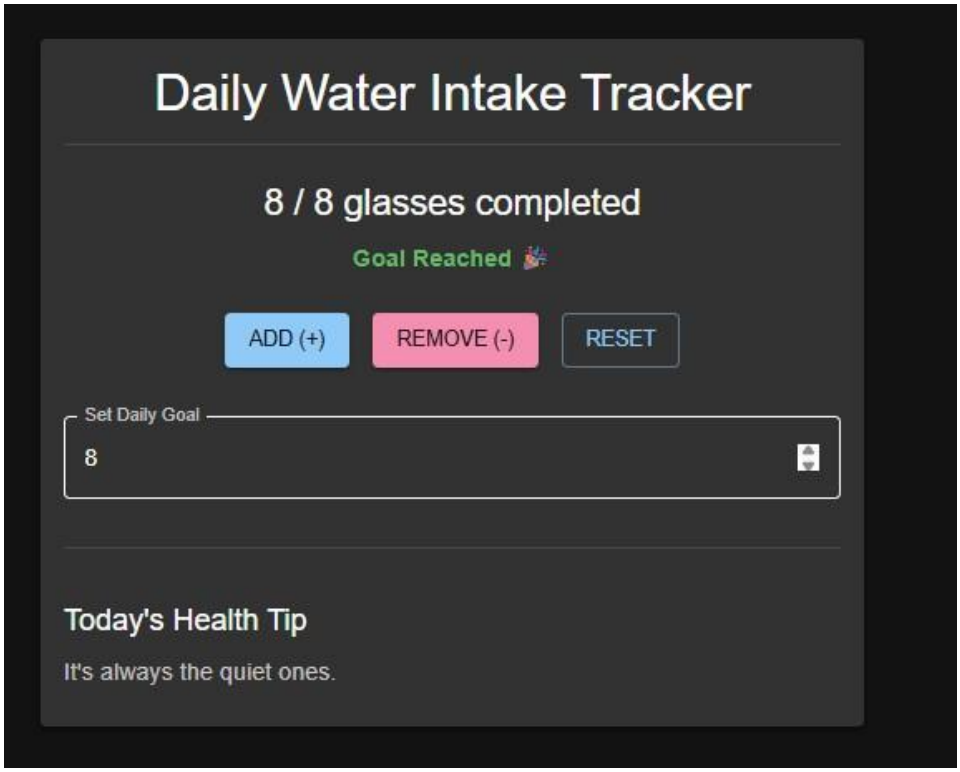
```jsx
import React from "react";
import ReactDOM from "react-dom/client";
import { BrowserRouter } from "react-router-dom";
import { ThemeProvider, createTheme, CssBaseline } from "@mui/material";
import App from "./App";

const darkTheme = createTheme({
  palette: {
    mode: "dark",
    primary: {
      main: "#90caf9",
    },
    secondary: {
      main: "#f48fb1",
    },
    background: {
      default: "#121212",
      paper: "#1e1e1e",
    },
  },
  typography: {
    fontFamily: "Roboto, sans-serif",
  },
});

ReactDOM.createRoot(document.getElementById("root")).render(
  <BrowserRouter>
    <ThemeProvider theme={darkTheme}>
      <CssBaseline />
      <App />
    </ThemeProvider>
  </BrowserRouter>
);
```

5. Output


EcoTrack Dashboard
DASHBOARD
WATER TRACKER
LOGOUT
Welcome to EcoTrack
This dashboard allows users to track their daily water intake, set hydration goals, and receive health tips.


Daily Water Intake Tracker
8 / 8 glasses completed
Goal Reached 🎉
ADD (+)
REMOVE (-)
RESET
Set Daily Goal
8
Today's Health Tip
It's always the quiet ones.

6. Learning Outcome :-

- Implemented protected routing to restrict unauthorized access.
- Designed a dark-themed professional UI using Material UI components.
- Managed application state using React Hooks (useState, useEffect).
- Persisted user data using browser localStorage.
- Integrated an external REST API to fetch daily health tips.
- Implemented loading and error handling states in asynchronous operations.